

Scalar

DEFINITION

A scalar is one number. In this course, scalars are real numbers, which can lie anywhere on the number line.

$x \in \mathbb{R}$ means “x is a real number.”

WORKED EXAMPLE

A temperature reading is 24.5 °C. Its numerical value, 24.5, is a scalar.

METAPHOR

One dial shows one number. Later, many dials will become a vector.

Variable

DEFINITION

A variable is a symbol whose value can vary within a specified set. An equation states a relationship between values.

$$y = 2x + 1$$

WORKED EXAMPLE

If $x = 3$, then $y = 2 \times 3 + 1 = 7$.

METAPHOR

A labeled empty box can hold different numbers while the rule stays the same.

Function

DEFINITION

A function assigns exactly one output to each allowed input. Its domain is the set of allowed inputs.

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$$

WORKED EXAMPLE

$f(3) = 9$ and $f(-3) = 9$. Different inputs may share an output.

METAPHOR

A machine follows one recipe for each input, even if two ingredients produce the same result.

Units

DEFINITION

A physical quantity is a numerical value multiplied by a unit. Only quantities with compatible dimensions can be added.

$$\text{distance} = \text{speed} \times \text{time}$$

WORKED EXAMPLE

3 metres/second $\times$ 4 seconds = 12 metres. Seconds cancel.

METAPHOR

Units are labels on measuring cups. Mixing litres with metres makes no meaningful total.

Vector

DEFINITION

A vector in $\mathbb{R}^n$ is an ordered list of n real numbers. Each position is a coordinate or feature.

$$x = (x_1, \dots, x_n); 2x = (2x_1, \dots, 2x_n)$$

WORKED EXAMPLE

If $x = (2, 3)$, then $2x = (4, 6)$. Order matters: $(2, 3) \neq (3, 2)$.

METAPHOR

A vector is a dashboard: each dial has its own fixed position.

Matrix

DEFINITION

A matrix is a rectangular array of numbers with m rows and n columns. Entry A_{ij} lies in row i , column j .

$$A \in \mathbb{R}^{m \times n}; A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

WORKED EXAMPLE

This 2×3 matrix has six entries. $A_{23} = 6$. Each inner bracket is one row.

METAPHOR

A spreadsheet with numbered rows and columns is a matrix when its cells are numbers.

Tensor

DEFINITION

In machine learning, a tensor organizes numbers along axes. Each entry has one index per axis. A vector has one axis, a matrix two, and a stack of matrices three.

$T \in \mathbb{R}^{2 \times 2 \times 2}$; T_{ijk} selects layer i , row j , column k

WORKED EXAMPLE

Layers: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$. $T_{212} = 6$: layer 2, row 1, column 2. Its shape is $(2, 2, 2)$.

METAPHOR

A book of number tables needs a page, row, and column to locate one number.

Summation

DEFINITION

Summation adds a specified sequence of terms. The symbol Σ says which index varies and where it starts and ends.

$$\sum_{i=1}^n x_i = x_1 + x_2 + \cdots + x_n$$

WORKED EXAMPLE

For $x = (2, 4, 6)$, $\sum_{i=1}^3 x_i = 12$. Their mean is $12/3 = 4$.

METAPHOR

Σ is an instruction to sweep the named numbers into one adding bucket.

Dot product

DEFINITION

The dot product of two equal-length vectors multiplies matching coordinates and adds the products.

$$x \cdot w = \sum_i x_i w_i$$

WORKED EXAMPLE

With $x = (2, 3)$ and $w = (4, 5)$, $x \cdot w = 8 + 15 = 23$.

METAPHOR

A shopping bill multiplies each quantity by its price, then adds the costs.

Matrix multiplication

DEFINITION

For A of shape $m \times n$ and B of shape $n \times p$, AB has shape $m \times p$. Each entry is a row-column dot product.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

WORKED EXAMPLE

$[[1, 2], [3, 4]] \times [[5], [6]] = [[17], [39]]$. First entry: $1 \times 5 + 2 \times 6 = 17$. Second: $3 \times 5 + 4 \times 6 = 39$.

METAPHOR

Each row submits a shopping list to each column of prices.

Transpose

DEFINITION

The transpose of a matrix exchanges rows and columns. It changes an $m \times n$ matrix into an $n \times m$ matrix.

$$(A^T)_{ij} = A_{ji}$$

WORKED EXAMPLE

If $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$, then $A^T = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

METAPHOR

Turn a row of books into a vertical stack without changing their order.

Linear transformation

DEFINITION

A linear transformation preserves vector addition and scalar multiplication. In finite dimensions, it can be represented by a matrix.

$$T(ax + by) = aT(x) + bT(y)$$

WORKED EXAMPLE

For $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $A(1, 2) = (2, 6)$. The first coordinate doubles; the second triples.

METAPHOR

Stretch a sheet by fixed factors along its axes, keeping the origin pinned.

Euclidean distance

DEFINITION

The Euclidean norm measures vector length. The distance between two vectors is the norm of their difference.

$$\|x\|_2 = \sqrt{(\sum_i x_i^2)}; d(x, y) = \|x - y\|_2$$

WORKED EXAMPLE

Between (0, 0) and (3, 4), distance = $\sqrt{9 + 16} = 5$.

METAPHOR

Stretch a straight string between two pins instead of following the grid streets.

Probability

DEFINITION

Probability measures how likely an event is on a scale from 0 to 1. Probabilities of all possible, nonoverlapping outcomes add to 1.

$$0 \leq P(A) \leq 1; \sum_i P(A_i) = 1$$

WORKED EXAMPLE

For a fair six-sided die, $P(\text{even}) = 3/6 = 0.5$.

METAPHOR

Divide one whole pie among the possible outcomes. Larger slices mean greater probability.

Conditional probability

DEFINITION

Conditional probability measures how likely A is when B is known to have occurred, provided $P(B) > 0$.

$$P(A \mid B) = P(A \cap B) / P(B)$$

WORKED EXAMPLE

Of four red balls, one is striped. $P(\text{striped} \mid \text{red}) = 1/4$.

METAPHOR

Once you enter the red-ball room, count only the balls inside that room.

Expected value

DEFINITION

The expected value of a discrete numerical outcome is its probability-weighted average.

$$E[X] = \sum_x x P(X = x)$$

WORKED EXAMPLE

A reward is 0 or 10 with equal probability. $E[X] = 0.5 \times 0 + 0.5 \times 10 = 5$.

METAPHOR

The balancing point of possible rewards can sit where no individual reward exists.

Variance

DEFINITION

Variance is the expected squared distance from the mean μ . Standard deviation σ is its square root.

$$\mu = E[X]; \text{Var}(X) = E[(X - \mu)^2]; \sigma = \sqrt{\text{Var}(X)}$$

WORKED EXAMPLE

Values 2 and 6 are equally likely. Mean = 4. Variance = $[(2-4)^2 + (6-4)^2]/2 = 4$. Standard deviation = $\sqrt{4} = 2$.

METAPHOR

The mean locates the center of the arrow hits; variance describes how widely they spread.

Exponential and logarithm

DEFINITION

An exponential raises a fixed base to a variable power. A logarithm gives the power needed to obtain a number. Natural logarithms use the base $e \approx 2.718$.

$$\log_b(b^x) = x, b > 0 \text{ and } b \neq 1; \ln(x) = \log_e(x)$$

WORKED EXAMPLE

$2^3 = 8$, so $\log_2(8) = 3$. Likewise, $e^2 \approx 7.389$, so $\ln(7.389) \approx 2$.

METAPHOR

An exponential climbs a multiplicative ladder; a logarithm tells you how many rungs you climbed.

Derivative

DEFINITION

The derivative is a function's instantaneous rate of change, defined by the limit of slopes over shrinking intervals.

$$f'(x) = \lim_{h \rightarrow 0} [f(x + h) - f(x)] / h$$

WORKED EXAMPLE

For $f(x) = x^2$: $f(3.01) - f(3) = 0.0601$. The slope over this interval is $0.0601/0.01 = 6.01$, close to $f'(3) = 6$.

METAPHOR

A speedometer measures the slope of your distance-versus-time journey at this instant.

Integral

DEFINITION

A definite integral is the limit of sums of a function's values times small interval widths. It measures signed accumulation.

$$\int_a^b f(t) dt = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(a + i\Delta t) \Delta t$$
$$\Delta t = (b - a)/n$$

WORKED EXAMPLE

A flow of 2 litres/second for 3 seconds delivers $\int_0^3 2 dt = 6$ litres.

METAPHOR

A derivative reads the tap's flow rate; an integral counts the water collected in the bucket.

Gradient

DEFINITION

For a scalar function of several variables, the gradient collects its partial derivatives. Each partial varies one coordinate while holding the others fixed.

$$\nabla L = (\partial L / \partial x, \partial L / \partial y)$$

WORKED EXAMPLE

If $L = x^2 + 3y^2$, then $\nabla L = (2x, 6y)$. At $(1, 2)$, it is $(2, 12)$.

METAPHOR

A hill's gradient is a local uphill compass, with one component for each direction.

Chain rule

DEFINITION

When one function feeds another, a change passes through both. The chain rule multiplies their local derivatives to find the derivative of the combined function.

If $y = f(g(x))$, then $dy/dx = f'(g(x)) g'(x)$

WORKED EXAMPLE

Let $u = 2x$ and $y = u^2$. At $x = 3$, $u = 6$. The two slopes are $du/dx = 2$ and $dy/du = 12$, so $dy/dx = 12 \times 2 = 24$.

METAPHOR

Linked gears multiply their speed ratios. A change travels through every gear.

Optimization

DEFINITION

An objective assigns a score to each choice. Optimization finds an allowed choice with the smallest or largest score. The allowed choices form the constraint set C .

$$x^* = \arg \min_{x \in C} L(x)$$

WORKED EXAMPLE

For $L(x) = (x - 3)^2$ and $C = [0, 2]$, $x^* = 2$ and $L(x^*) = 1$.

METAPHOR

Find the lowest campsite inside the fence. A lower valley outside it is unavailable.

Gradient descent

DEFINITION

Gradient descent updates a variable or parameter vector in the direction opposite its gradient. The learning rate η sets the step size. A sufficiently small step can lower the loss.

$$\theta^+ = \theta - \eta \nabla L(\theta), \eta > 0$$

WORKED EXAMPLE

For $L(\theta) = \theta^2$, $\theta = 3$ and $\eta = 0.1$: $\theta^+ = 3 - 0.1 \times 6 = 2.4$. The loss falls from 9 to $2.4^2 = 5.76$.

METAPHOR

Feel the ground's slope, take a short downhill step, then measure again.

Dataset and features

DEFINITION

A dataset collects examples. Features describe each example numerically. In supervised learning, each input x is paired with a target answer y .

$$D = \{(x_i, y_i)\}_{i=1}^N$$

WORKED EXAMPLE

Two rooms have features (length, width): (2, 3) and (4, 3) metres. $X = \begin{bmatrix} 2 & 3 \\ 4 & 3 \end{bmatrix}$ has two rows and two features per row.

METAPHOR

Features are the clues on a detective's case card; the target is the answer being learned.

Linear regression

DEFINITION

Linear regression predicts a numerical target using a weighted sum of features plus an intercept.

$$\hat{y} = w \cdot x + b$$

WORKED EXAMPLE

A toy taxi fare model is $\hat{y} = 2x + 3$ dollars for x kilometres. At 5 km: $\hat{y} = 2 \times 5 + 3 = \13 . The intercept is the starting fare.

METAPHOR

A recipe assigns each ingredient a fixed contribution, then adds a fixed starting amount.

Mean squared error

DEFINITION

Mean squared error is the average squared difference between predictions and targets. It penalizes larger errors more strongly.

$$\text{MSE} = (1/N) \sum_i (\hat{y}_i - y_i)^2$$

WORKED EXAMPLE

Predictions (2, 5), targets (3, 3): $\text{MSE} = (1 + 4)/2 = 2.5$.

METAPHOR

Large misses get disproportionately heavier penalty weights than small misses.

Supervised learning

DEFINITION

Supervised learning uses examples with known answers to learn a rule for new inputs. A loss measures prediction error; training adjusts the rule to reduce that loss.

Training objective: minimize $(1/N) \sum_i \ell(f_{\theta}(x_i), y_i)$
 θ = model parameters; ℓ = loss for one example

WORKED EXAMPLE

Inputs 1, 2, 3 paired with targets 2, 4, 6 suggest $f(x) = 2x$; it predicts 8 for input 4.

METAPHOR

A coach gives both a problem and a worked answer so a learner can improve its rule.

Parameter and hyperparameter

DEFINITION

A parameter is fitted during learning. A hyperparameter configures the model or training procedure rather than being fitted by that procedure.

$$\hat{y} = wx + b; w^+ = w - \eta \partial L / \partial w$$

WORKED EXAMPLE

In this update, $w = 2$ is a parameter; the chosen learning rate $\eta = 0.01$ is a hyperparameter.

METAPHOR

The learner adjusts recipe quantities; the teacher chooses how large each adjustment may be.

Training, validation and test

DEFINITION

Training data fit parameters; validation data guide model choices; a held-out test estimates performance after those choices are fixed.

$D = D_{\text{train}} \cup D_{\text{val}} \cup D_{\text{test}}$; sets are disjoint

WORKED EXAMPLE

A 100-well study might use 70 wells to train, 15 to choose settings, and 15 for final testing.

METAPHOR

Practice exercises, a mock exam, and a sealed final exam have different jobs.

Data leakage

DEFINITION

Data leakage occurs when model development uses information that would be unavailable in the intended prediction setting.

For new-well testing: train wells n test wells = $\emptyset$

WORKED EXAMPLE

Training uses a well segment at depths 100, 101, 102 m.

Testing uses 101, 102, 103 m from the same well. Two of three measurements are shared.

METAPHOR

An exam is misleading when the practice sheet contains copies of its questions.

Normalization

DEFINITION

Normalization rescales data to a chosen reference scale.

Standardization, one common method, subtracts the mean μ and divides by the standard deviation σ .

Standardization: $z = (x - \mu)/\sigma, \sigma > 0$

WORKED EXAMPLE

Training data give $\mu = 100$ and $\sigma = 20$. A new value $x = 140$ becomes $z = 2$.

METAPHOR

Convert different measuring rulers into a common “distance from typical” scale.

Artificial neuron

DEFINITION

An artificial neuron multiplies inputs by weights, adds a bias, then applies an output rule called an activation function. The bias shifts the sum by a constant.

$$a = \phi(w \cdot x + b)$$

WORKED EXAMPLE

With $x = (2, 1)$, $w = (3, -2)$, $b = 1$: $z = 3 \times 2 - 2 \times 1 + 1 = 5$.
The activation $\max(0, z)$ gives $a = 5$.

METAPHOR

A tiny decision station combines weighted signals, then applies one output rule.

Sigmoid

DEFINITION

The logistic sigmoid maps a real input smoothly into the open interval $(0, 1)$. Its slope becomes small at extreme inputs.

$$\sigma(z) = 1/(1 + e^{-z}); \sigma'(z) = \sigma(z)(1 - \sigma(z))$$

WORKED EXAMPLE

$\sigma(-2) \approx 0.119$, $\sigma(0) = 0.5$, $\sigma(2) \approx 0.881$. At zero, the slope is $0.5 \times (1 - 0.5) = 0.25$.

METAPHOR

A dimmer gradually opens from almost off to almost fully on.

ReLU

DEFINITION

The rectified linear unit keeps positive inputs and sets negative inputs to zero. It is a piecewise-linear activation.

$$\text{ReLU}(z) = \max(0, z)$$

WORKED EXAMPLE

$\text{ReLU}(-3) = 0$; $\text{ReLU}(2) = 2$. Its slope is 0 below zero and 1 above zero.

METAPHOR

A one-way gate blocks negative signals and lets positive ones pass unchanged.

Forward pass

DEFINITION

A forward pass computes a model's output using its current weights. Intermediate results follow a computation graph: the sequence of calculations connecting input to output.

Example model: $\hat{y} = wx$; training loss: $L = (\hat{y} - y)^2$

WORKED EXAMPLE

With $x = 2$ and $w = 1$, predict $\hat{y} = 2$. If the target is $y = 6$, the loss is $(2 - 6)^2 = 16$. The weight is still 1.

METAPHOR

Run ingredients through the workshop to see the current product before changing the machines.

Backpropagation

DEFINITION

Backpropagation applies the chain rule backward through a model to compute how its loss changes with each weight. It calculates gradients; a separate update changes the weights.

$$\partial L / \partial w = (\partial L / \partial \hat{y})(\partial \hat{y} / \partial w) = 2(\hat{y} - y)x$$

WORKED EXAMPLE

For $\hat{y} = wx$, $x = 2$, $w = 1$, $y = 6$: $\hat{y} = 2$. The local derivatives are $2(2 - 6) = -8$ and $x = 2$. Multiply: $\partial L / \partial w = -16$.

METAPHOR

Trace a faulty product backward through the workshop to measure each station's contribution.

Mini-batch gradient descent

DEFINITION

Mini-batch stochastic gradient descent estimates a training gradient using a subset B of examples, then updates the parameters.

$$g^B = \sum_{i \in B} \nabla \ell_i / \text{size}(B); \theta^+ = \theta - \eta g^B$$

WORKED EXAMPLE

Two examples give gradients 2 and 6, averaging to 4. With $\theta = 1$ and $\eta = 0.1$, the update is $\theta^+ = 1 - 0.1 \times 4 = 0.6$.

METAPHOR

Taste a small spoonful of soup before adjusting the whole pot.

Epoch

DEFINITION

An epoch is one pass through a training dataset. A step is one parameter update, often using a mini-batch.

$\text{steps per epoch} = \text{ceiling}(N / B)$

WORKED EXAMPLE

With 100 examples and batch size 20, one epoch contains 5 steps; three epochs contain 15.

METAPHOR

An epoch is one complete lap; mini-batches are the strides within it.

Deep neural network

DEFINITION

A deep neural network passes data through multiple hidden layers before producing its output. Each layer learns weights and biases, usually followed by a nonlinear activation.

$$h_0 = x; h_l = \phi_l(W_l h_{l-1} + b_l)$$

WORKED EXAMPLE

At $x = 2$: $h_1 = \text{ReLU}(2x-1) = 3$; $h_2 = \text{ReLU}(h_1-1) = 2$; output $y = 3h_2 = 6$. These are two hidden layers and an output layer.

METAPHOR

A workshop has successive stations: each transforms what the previous station produced.

Overfitting

DEFINITION

Overfitting occurs when a model learns training-specific details that do not generalize to the intended new data.

generalization gap = test error – training error

WORKED EXAMPLE

A model has 1% training error but 20% test error. Its measured gap is 19 percentage points.

METAPHOR

Memorizing the practice exam can look like understanding until the questions change.

Regularization

DEFINITION

Regularization limits how a model fits its training data to help it generalize. L2 regularization adds a penalty for large weights to the prediction loss.

$$L_{\text{total}} = L_{\text{data}} + \lambda \|w\|_2^2, \lambda \geq 0$$

WORKED EXAMPLE

If $L_{\text{data}} = 2$, $w = (3, 4)$, and $\lambda = 0.1$, total loss = $2 + 0.1 \times 25 = 4.5$.

METAPHOR

Charge rent for very large knobs so the model needs a good reason to turn them far.

Softmax

DEFINITION

Softmax converts a vector of real scores, called logits, into positive numbers that sum to one.

$$p_i = \exp(z_i) / \sum_j \exp(z_j)$$

WORKED EXAMPLE

Logits (0, ln 3) give exponentials (1, 3), so probabilities are (0.25, 0.75).

METAPHOR

Convert competing bids into shares of one whole budget.

Cross-entropy

DEFINITION

Cross-entropy scores predicted probabilities against target probabilities. It is the average negative log probability of the target outcomes. Lower scores reward better predictions.

$$H(q, p) = -\sum_i q_i \ln p_i$$

WORKED EXAMPLE

The correct class is cat. Predicting $p(\text{cat}) = 0.9$ gives $-\ln(0.9) \approx 0.105$. Predicting 0.1 gives $-\ln(0.1) \approx 2.303$, a much larger loss.

METAPHOR

A confident wrong answer pays a much bigger surprise bill than a cautious one.

Self-supervised learning

DEFINITION

Self-supervised learning constructs prediction targets from the data themselves, without requiring a separately supplied label for every example.

$$L = \ell(f_{\theta}(\text{mask}(x)), x_{\text{hidden}})$$

WORKED EXAMPLE

Hide the middle value in (2, 4, 6). The model sees (2, ?, 6) and predicts 5. The hidden target is 4, so squared error = $(5-4)^2 = 1$.

METAPHOR

Cover a word in your own book, guess it, then uncover the answer.

Embedding

DEFINITION

An embedding represents an object as a vector. Learned embeddings adjust these coordinates so distances or directions can capture relationships useful to a task.

$$f: \text{objects} \rightarrow \mathbb{R}^d; z = f(x)$$

WORKED EXAMPLE

Toy word vectors: cat = (1, 0), dog = (0.9, 0.1), rock = (0, 1).
Cat is closer to dog (distance ≈ 0.14) than rock (≈ 1.41).

METAPHOR

A map gives complicated places compact coordinates, while preserving only some relationships.

Cosine similarity

DEFINITION

Cosine similarity compares directions of two nonzero vectors by dividing their dot product by their lengths.

$$\cos(x, y) = (x \cdot y) / (\|x\|_2 \|y\|_2)$$

WORKED EXAMPLE

For $x = (1, 0)$, $y = (2, 0)$, similarity $= 2 / (1 \times 2) = 1$, despite different lengths.

METAPHOR

Two arrows can point the same way even when one is twice as long.

Principal component analysis

DEFINITION

PCA subtracts the data's mean, then finds perpendicular directions that preserve the most variance. Projecting onto the leading directions gives fewer coordinates per example.

First direction: maximize $E[(u \cdot x)^2]$, with $\|u\|_2 = 1$

x is centered: its mean is zero

WORKED EXAMPLE

Points $(-1, -1)$ and $(1, 1)$ lie on a diagonal. Projecting onto $u = (1, 1)/\sqrt{2}$ gives $-\sqrt{2}$ and $\sqrt{2}$. One coordinate locates each point.

METAPHOR

Rotate a flashlight until a one-dimensional shadow preserves the object's greatest spread.

Autoencoder

DEFINITION

An autoencoder learns to reconstruct its input. Its encoder produces a code z ; its decoder rebuilds the input from z . Training reduces the reconstruction error.

$$z = f_{\theta}(x); \hat{x} = g_{\phi}(z); L = \text{MSE}(x, \hat{x})$$

Training adjusts encoder weights θ and decoder weights ϕ .

WORKED EXAMPLE

With toy weights: $x = (2, 5)$, encoder $z = x_1 = 2$, decoder $\hat{x} = (z, 2z) = (2, 4)$. Reconstruction loss $= (0^2 + 1^2)/2 = 0.5$.

METAPHOR

Pack a suitcase, then unpack it and check which belongings survived.

Latent space and bottleneck

DEFINITION

A latent vector encodes an input inside a model. Latent space contains its possible codes. A bottleneck limits the size or information capacity of that representation.

Undercomplete bottleneck: $x \in \mathbb{R}^n \rightarrow z \in \mathbb{R}^d, d < n$

WORKED EXAMPLE

Pairs (2, 4), (3, 6), (5, 10) can use one-number codes 2, 3, 5. Decoder ($z, 2z$) restores every pair. Input (2, 5) breaks that pattern.

METAPHOR

A narrow doorway forces a traveler to choose what to carry through.

Bit and byte

DEFINITION

A bit has two possible values, 0 or 1. A byte contains eight bits and can encode 256 different bit patterns.

n bits encode 2^n patterns; 1 byte = 8 bits

WORKED EXAMPLE

Two bits have four patterns: 00, 01, 10, 11. An agreed binary encoding maps them to 0, 1, 2, 3. Eight bits give $2^8 = 256$ patterns.

METAPHOR

Bits are tiny switches; eight switches form a byte-sized combination lock.

Numerical precision

DEFINITION

Numerical precision describes how finely a number format represents values. Limited bits require rounding and impose a finite range.

absolute rounding error = $\text{abs}(x - \text{rounded value})$

WORKED EXAMPLE

Rounding 1.234 to two decimal places gives 1.23, an error of 0.004.

METAPHOR

A ruler marked only in centimetres cannot record every millimetre exactly.

Rate-distortion tradeoff

DEFINITION

Lossy compression balances representation size, or rate R , against reconstruction error, or distortion D .

Choose a code to minimize $D + \lambda R$

D = reconstruction error; R = bits per input

WORKED EXAMPLE

Code A: $D = 1$, $R = 20$ bits. Code B: $D = 2$, $R = 5$ bits. At $\lambda = 0.1$, scores are 3 and 2.5. B wins by accepting more error for fewer bits.

METAPHOR

Packing lighter is useful only if the missing items are worth the saved luggage space.

Vector quantization

DEFINITION

Vector quantization maps a vector to one of a finite set of codebook vectors. Nearest-neighbor quantization chooses the closest entry.

$$k^* = \arg \min_k \|z - e_k\|_2^2; z_q = e_{k^*}$$

WORKED EXAMPLE

For $z = (1, 2)$, codebook entries $e_1 = (0, 2)$ and $e_2 = (3, 2)$ are 1 and 2 units away. Choose e_1 , and store its index 1.

METAPHOR

Choose the nearest paint swatch instead of describing an infinitely precise color.

Token and vocabulary

DEFINITION

A token is a discrete symbol identified within a vocabulary, a finite set of allowed symbols. It need not be a word.

token $k \in V = \{1, \dots, K\}$; embedding = $E[k]$

WORKED EXAMPLE

Codebook entry 6 is the vector (1, 2). Token 6 identifies that entry; a lookup retrieves (1, 2). A vocabulary of 8 entries needs 3 bits per fixed-length ID.

METAPHOR

A coat-check ticket is a discrete ID; the numbered ticket is not the coat itself.

Vector-quantized autoencoder

DEFINITION

A VQ-VAE is an autoencoder whose decoder receives vectors selected from a learned codebook. Each continuous encoder output is replaced by its nearest codebook entry.

$x \rightarrow z \rightarrow \text{nearest code } z_q \rightarrow \hat{x}$

Straight-through estimate: $\partial L / \partial z \approx \partial L / \partial z_q$

WORKED EXAMPLE

For $x = (2, 5)$, the encoder gives $z = 1.9$. Codebook $\{0, 2\}$ selects 2. Decoder $(z_q, 2z_q)$ gives $(2, 4)$, with $\text{MSE} = (0+1)/2 = 0.5$.

METAPHOR

Pack with a catalog of standard boxes, and improve both the packing method and box catalog.

Sequence and autocorrelation

DEFINITION

A sequence is an ordered list. Autocorrelation compares its values with the same sequence shifted by k positions, called the lag. It measures how values vary together across that gap.

$$r_k = \sum_{i=1}^{n-k} (x_i - \mu)(x_{i+k} - \mu) / \sum_{i=1}^n (x_i - \mu)^2$$

μ = sample mean; r_k is a finite-sample estimate

WORKED EXAMPLE

For (1, 3, 1, 3), mean = 2 and deviations = (−1, 1, −1, 1). At lag 1: $r_1 = (-1-1-1)/(1+1+1+1) = -0.75$. Neighbors alternate.

METAPHOR

Loud, soft, loud, soft: shift the rhythm by one beat and loud lines up with soft.

Autoregressive language model

DEFINITION

An autoregressive language model assigns a sequence probability by predicting each next token conditioned on the previous tokens.

$$P(t_1, \dots, t_n) = \prod_i P(t_i \mid t_1, \dots, t_{i-1})$$

WORKED EXAMPLE

If $P(A) = 0.5$ and $P(B \text{ given } A) = 0.2$, then $P(A, B) = 0.1$.

METAPHOR

Write a sentence one tile at a time, choosing each tile from the tiles already placed.

Positional encoding

DEFINITION

Positional encoding gives a model token order or location. The additive method adds a position vector to each embedding. Other methods alter attention.

Additive version: $h_i = E[\text{token}_i] + p_i$

WORKED EXAMPLE

The same token has vector (1, 2). At position 1, add (0.1, 0): (1.1, 2). At position 2, add (0, 0.1): (1, 2.1). Its position now changes the input.

METAPHOR

Page numbers let you distinguish repeated words appearing in different places in a book.

Scaled dot-product attention

DEFINITION

Each query scores a set of keys. Softmax turns the scores into weights. Attention returns a weighted sum of the corresponding values, letting a token use information from other tokens.

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T/\sqrt{d_k})V$$

WORKED EXAMPLE

$q = 1$, keys = (0, ln 3), values = (2, 6), $d_k = 1$. Scores (0, ln 3) give weights (0.25, 0.75). Output = $0.25 \times 2 + 0.75 \times 6 = 5$.

METAPHOR

A question asks several reference cards how relevant they are, then blends their answers.

Multi-head attention

DEFINITION

Multi-head attention runs several attention calculations using different learned matrix transformations. Their outputs are joined and mixed by another learned matrix.

$$\text{MultiHead} = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

WORKED EXAMPLE

For values (2, 6), weights (0.25, 0.75) give 5; weights (0.75, 0.25) give 3. Join the heads as (5, 3). Output weights (0.5, 0.5) give 4.

METAPHOR

Several readers inspect the same passage with different questions, then combine their notes.

Causal mask

DEFINITION

A causal mask blocks attention to future tokens. It adds $-\infty$ to their attention scores, so softmax assigns them zero weight. Each position can use only its present and past.

$$M_{ij} = 0 \text{ if } j \leq i; M_{ij} = -\infty \text{ if } j > i$$

WORKED EXAMPLE

At position 2, start with scores (0, 0, 0). Mask position 3: (0, 0, $-\infty$). Softmax gives (0.5, 0.5, 0): the future contributes nothing.

METAPHOR

Fold the exam paper so the answer to the next question remains hidden.

Transformer

DEFINITION

A Transformer uses attention to mix information across tokens and feed-forward layers to transform each token. Residual additions and layer normalization help train many such layers.

One block: $u = \text{LN}(h + \text{Attention}(h))$

$h' = \text{LN}(u + \text{FFN}(u))$; LN normalizes each token's features

WORKED EXAMPLE

Illustrating the first stage: $h = (1, 3)$, attention output = $(1, 1)$, sum = $(2, 4)$. Mean 3 and standard deviation 1 give approximately $(-1, 1)$.

METAPHOR

Revise a shared notebook repeatedly, preserving the old notes while adding improvements.

Fine-tuning

DEFINITION

Fine-tuning continues optimization of some or all parameters of a pretrained model using a new dataset or objective.

$$\theta^+ = \theta_0 - \eta \nabla L_{\text{new}}(\theta_0); \theta_0 \text{ is pretrained}$$

WORKED EXAMPLE

A trainable weight starts at 0.5. With gradient 2 and $\eta = 0.01$, one update gives 0.48.

METAPHOR

Teach an experienced musician a new repertoire without starting musical training from zero.

Retrieval-augmented generation

DEFINITION

Retrieval-augmented generation finds stored information relevant to a question and gives it to a language model as context for its answer.

$$c = \text{Retrieve}(q, D); \text{answer} \sim P_{\theta}(\cdot \mid q, c)$$

WORKED EXAMPLE

Question: “How deep is Well A?” Retrieve record 17: “Well A, total depth 2,400 m.” The answer reports 2,400 m and points to record 17.

METAPHOR

An open-book exam lets you consult selected pages before composing an answer.

API and SDK

DEFINITION

An application programming interface defines how software components interact. A software development kit supplies libraries and tools for developers to use those interfaces.

Example API contract: `mean(list of numbers)` → number

WORKED EXAMPLE

Calling `mean([2, 4, 6])` returns 4. An SDK might provide a function that sends this request to a service and reads its response.

METAPHOR

An API is a socket's specification; an SDK is the toolkit for building a compatible plug.

Tool calling

DEFINITION

Tool calling lets a model request a structured operation from external software, then use the returned result. The software defines the operation's semantics.

`request = {tool, arguments}; result = tool(arguments)`

WORKED EXAMPLE

A model requests `mean([10, 20, 30])`; the calculator returns 20, which the answer reports.

METAPHOR

A coordinator asks a calculator to calculate instead of guessing the result from memory.

State, observation and action

DEFINITION

A state contains enough information to predict the system's next-state distribution given an action. An observation is a measurement of the system. An action is an input we choose.

$$s_{t+1} = F(s_t, a_t); o_t = H(s_t) + \text{noise}$$

WORKED EXAMPLE

In a simple tank model, state $s = 10$ litres. A noisy sensor reports observation $o = 9.8$ litres. Choosing to add 3 litres is action $a = 3$.

METAPHOR

The whole room is the state; a keyhole view is the observation; turning the handle is an action.

Dynamical system

DEFINITION

A dynamical system has a state that changes over time. Its transition rule describes how the next state depends on the current state, any chosen action, and disturbances.

$$s_{t+1} = F(s_t, a_t, \xi_t)$$

s = state; a = action; ξ = disturbance; t = time step

WORKED EXAMPLE

A tank holds 10 litres. Add 3 and drain 2 during one step: next volume = $10 + 3 - 2 = 11$.

METAPHOR

The present is a frame in a movie; dynamics specify how the next frame can unfold.

Markov decision process

DEFINITION

An MDP describes states, actions, rewards and transition probabilities. Given the current state and action, the next-state distribution does not depend on earlier history.

$$P(s_{t+1} \mid \text{history}, a_t) = P(s_{t+1} \mid s_t, a_t)$$

WORKED EXAMPLE

In state “low,” choosing “fill” reaches “safe” with probability 0.9 and stays “low” with probability 0.1. Suppose reaching “safe” earns reward 1, otherwise 0.

METAPHOR

A board-game position contains everything needed to determine the next move's possibilities.

Reward and return

DEFINITION

A reward scores one outcome. Return adds rewards over several steps. A discount factor γ gives less weight to later rewards when $0 \leq \gamma < 1$.

For H remaining steps: $G_t = \sum_{k=0}^{H-1} \gamma^k r_{t+k}$
 r_t is the reward from action a_t ; $0 \leq \gamma \leq 1$

WORKED EXAMPLE

For three rewards (2, 0, 10) and $\gamma = 0.5$, return = $2 + 0 + 0.25 \times 10 = 4.5$.

METAPHOR

Immediate reward is one coin; return is the discounted value of the entire trail of coins.

Policy

DEFINITION

A policy specifies how an agent chooses actions. A stochastic policy gives an action probability for each state or available information.

$$\pi(a \mid s) \geq 0; \sum_a \pi(a \mid s) = 1$$

WORKED EXAMPLE

In a low tank, $\pi(\text{fill}) = 0.8$ and $\pi(\text{wait}) = 0.2$. A deterministic policy might always fill.

METAPHOR

A playbook says what to do when a particular situation appears.

Value function

DEFINITION

A state-value function is the expected return starting in a state and following a specified policy. An action-value also fixes the first action.

$$V_{\pi}(s) = E_{\pi}[G_t \mid s_t = s]; Q_{\pi}(s, a) = E_{\pi}[G_t \mid s, a]$$

WORKED EXAMPLE

A policy yields return 10 with probability 0.6 and 0 otherwise, so $V_{\pi}(s) = 6$.

METAPHOR

Value is a forecast of the whole journey's worth, not the reward on the next step.

Bellman equation

DEFINITION

The Bellman equation decomposes a policy's value into expected immediate reward plus discounted value of the next state.

$$V_{\pi}(s) = E_{\pi}[r_t + \gamma V_{\pi}(s_{t+1}) \mid s_t = s]$$

WORKED EXAMPLE

Reward now is 2. Next-state value is 10 or 0, equally likely, under the policy. At $\gamma = 0.9$: $V = 2 + 0.9 \times (0.5 \times 10 + 0.5 \times 0) = 6.5$.

METAPHOR

A journey's worth is this leg's reward plus what remains beyond the next stop.

Exploration and exploitation

DEFINITION

Exploitation chooses what currently appears best. Exploration gathers information by trying alternatives that may improve future decisions.

ϵ -greedy: explore with probability ϵ ; otherwise choose $\arg \max_a Q(s, a)$

WORKED EXAMPLE

With $\epsilon = 0.1$ over 100 decisions, expect about 10 exploration-branch choices, not exactly 10.

METAPHOR

Return to your favorite food stall, but sometimes try another to discover a better one.

Q-learning

DEFINITION

Q-learning updates an action-value estimate toward a target using observed reward and the largest estimated next-state action value.

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_b Q(s', b) - Q(s, a)]$$

WORKED EXAMPLE

$Q = 2$, reward = 1, $\gamma = 0.5$, best next $Q = 6$. Target = $1 + 0.5 \times 6 = 4$. With $\alpha = 0.25$, new $Q = 2 + 0.25 \times (4 - 2) = 2.5$.

METAPHOR

Revise a travel guide whenever a journey reveals a better estimate of the route's payoff.

Policy gradient

DEFINITION

Policy-gradient methods adjust a policy's parameters to increase expected return. They add a step along the gradient, which points toward local improvement.

$$J(\theta) = E\pi_{\theta}[G]; \theta^+ = \theta + \eta \nabla J(\theta)$$

WORKED EXAMPLE

Choose A with probability p . Rewards: A gives 4, B gives 1. $J(p) = 4p + (1-p) = 1+3p$, so $J'(p) = 3$. With $\eta = 0.1$: $p^+ = 0.5 + 0.1 \times 3 = 0.8$.

METAPHOR

Strengthen decisions that led to better outcomes, using measured credit rather than copying every action.

Model-based RL

DEFINITION

Model-based RL uses a model of environment changes or rewards to help choose actions. It can plan with this model or learn from simulated experience.

$\hat{s}_{t+1} = \hat{F}(s_t, a_t)$; choose actions using predicted returns

WORKED EXAMPLE

A learned tank model predicts that “fill” yields return 7 and “wait” yields 5. Choose “fill,” observe the real outcome, and use it to check or improve the model.

METAPHOR

Practice routes in a flight simulator before flying the aircraft.

Safety constraint

DEFINITION

A safety constraint restricts allowed states or actions. A constrained controller optimizes only among choices satisfying its stated limits.

maximize $\text{reward}(a)$, subject to $g(a) \leq 0$

WORKED EXAMPLE

If a valve setting must satisfy $a \leq 5$, a proposed setting of 7 is rejected even if its predicted reward is high.

METAPHOR

A game's boundary line is a rule, not a small negative score you may pay to ignore.

Model predictive control

DEFINITION

MPC plans several actions using a model, executes the first, then replans using new measurements. It minimizes a cost over a fixed number of future steps.

Minimize $\sum_{k=0}^{H-1} c(s_{k+1}, a_k)$, with $s_{k+1} = F(s_k, a_k)$

H = planning horizon; enforce state and action limits

WORKED EXAMPLE

A tank has 10 L, target 14 L, drain 2 L/step. Plan inflows (6, 2) L. Apply 6 now. If the tank actually reaches 13 L, replan the next inflow as 3 L.

METAPHOR

A navigator redraws the route at every junction as traffic changes.

RLHF

DEFINITION

Reinforcement learning from human feedback (RLHF) often learns a reward model from human preferences, then trains a policy to earn higher scores.

For a fixed question: $J(\theta) = \sum_y \pi_\theta(y) r(y)$

$\pi_\theta(y)$ = probability of answer y ; $r(y)$ = learned reward

WORKED EXAMPLE

People prefer answer A to B. Suppose learned rewards are 2 and 0. Raising $P(A)$ from 0.5 to 0.75 raises expected reward from 1 to 1.5.

METAPHOR

A judge compares performances; the learner improves toward the judge's scoring pattern.

CPU

DEFINITION

A central processing unit (CPU) executes program instructions: arithmetic, comparisons, and decisions about what to run next. A computer can use several CPU cores at once.

Example program: $\text{total} \leftarrow \text{total} + x_i$; $\text{mean} = \text{total}/N$

WORKED EXAMPLE

To average (2, 4, 6), a program starts $\text{total} = 0$. Add each value: 2, then 6, then 12. Divide by 3 to return 4.

METAPHOR

A versatile workshop supervisor handles varied jobs and directs specialist machines.

GPU

DEFINITION

A graphics processing unit (GPU) performs many similar calculations in parallel. Large matrix operations supply repeated arithmetic to keep its execution units busy.

Independent elementwise work: $y_i = 2x_i$

WORKED EXAMPLE

For $x = (1, 2, 3, 4)$, compute $y = (2, 4, 6, 8)$. Each output can be calculated independently. The same pattern scales to millions of entries.

METAPHOR

A large team folds many identical paper cranes at once.

Tensor core

DEFINITION

A Tensor Core is a specialized circuit inside an NVIDIA GPU that multiplies small matrix blocks and adds another matrix. Supported number formats depend on the GPU generation.

$$D = AB + C$$

WORKED EXAMPLE

$A = \begin{bmatrix} 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 5 \end{bmatrix}$. Multiply: $AB = \begin{bmatrix} 1 \times 3 + 2 \times 4 \end{bmatrix} = \begin{bmatrix} 11 \end{bmatrix}$. Add C: $D = \begin{bmatrix} 16 \end{bmatrix}$.

METAPHOR

A factory station stamps whole groups of arithmetic steps using a specialized mold.

CUDA

DEFINITION

CUDA is NVIDIA's platform for programming its GPUs. A kernel is a function run by many logical workers called threads. Threads have indices and are organized into blocks.

For a 1D launch: $i = \text{block index} \times B + \text{thread index}$
 $B = \text{threads per block}$; indices start at 0

WORKED EXAMPLE

With $B = 4$, block 1 and thread 2 give $i = 1 \times 4 + 2 = 6$. That thread can compute $y_6 = 2x_6$. Eight such blocks launch $8 \times 4 = 32$ threads.

METAPHOR

CUDA is the instruction system that assigns numbered tasks to a parallel workshop.

Memory capacity

DEFINITION

Memory capacity is the amount of information a device can hold. Model execution needs space for weights, intermediate values, and working buffers.

$\text{weight bytes} = \text{parameter count} \times \text{bits per parameter} / 8$

WORKED EXAMPLE

1 billion parameters at 16 bits use 2 billion bytes, or 2 GB, for weights alone.

METAPHOR

A kitchen counter limits how many ingredients and partly prepared dishes fit at once.

Memory bandwidth

DEFINITION

Memory bandwidth is the amount of data transferred per unit time. It can limit speed even when computation units are fast.

$\text{transfer time} \geq \text{bytes moved} / \text{peak bytes per second}$

WORKED EXAMPLE

Moving 10 GB through a 100 GB/second channel takes at least 0.1 seconds under ideal conditions.

METAPHOR

A giant pantry does not help a cook if ingredients arrive through a narrow serving hatch.

H100

DEFINITION

The NVIDIA H100 is a Hopper-generation data-center GPU. Its variants differ; the H100 SXM specification includes 80 GB of HBM3 memory.

weights-only capacity estimate: $80 \text{ GB} / 2 \text{ bytes} = 40 \text{ billion values}$

WORKED EXAMPLE

A 10-billion-parameter model uses about 20 GB at 16 bits for weights, leaving nominal space for other needs.

METAPHOR

An H100 is a particular workshop machine, with a specified counter size and arithmetic tools.

Compute cluster

DEFINITION

A compute cluster is a set of networked computers working together. Distributed training may divide data, model parameters, or both across devices.

Simple model: $\text{time} = \text{serial work} + \text{parallel work}/n$
 n = equal workers; communication adds further time

WORKED EXAMPLE

An 80-second job has 20 seconds of serial work and 60 divisible seconds. Four workers need at least $20 + 60/4 = 35$ seconds in this model.

METAPHOR

Eight kitchens can cook together, but they still need roads to exchange ingredients and results.

TSMC and chip fabrication

DEFINITION

TSMC is a semiconductor foundry: it manufactures chips designed by customers. Circuits are formed on a silicon wafer, then separated into individual chips called dies.

$\text{yield} = \text{usable dies} / \text{manufactured dies}$

WORKED EXAMPLE

In a hypothetical batch of 100 dies, 80 passing inspection gives 80% yield. This is a teaching example, not TSMC production data.

METAPHOR

A foundry is the factory that fabricates another company's machine design.

Well log and LAS

DEFINITION

A well log records measurements indexed by depth. Log ASCII Standard, LAS, is a text format for exchanging well-log curves and metadata.

curve $x(d)$; samples = $\{(d_i, x_i)\}$

WORKED EXAMPLE

A toy temperature log records (100 m, 30 °C) and (101 m, 30.2 °C). Temperature rises 0.2 °C over 1 m. LAS stores curve names and units too.

METAPHOR

A well log is a vertical diary of instrument readings through the ground.

Windowing and missing data

DEFINITION

Windowing divides an ordered signal into local segments. A mask records which entries are observed so absent values are not mistaken for measurements.

$$W_i = (x_i, \dots, x_{i+l-1}); m_j \in \{0, 1\}$$

WORKED EXAMPLE

Sequence (2, 4, missing, 6) gives length-3 windows (2, 4, missing) and (4, missing, 6). The second has mask (1, 0, 1): its middle value is absent.

METAPHOR

Cut a long film into scenes, while marking damaged frames rather than pretending they show darkness.

Provenance and uncertainty

DEFINITION

Provenance records a result's origin and transformations.

Uncertainty describes what is not known about its value, using a stated method or bound.

If $\text{abs}(\text{error}) \leq \delta$: true value $\in [\text{reading} - \delta, \text{reading} + \delta]$

WORKED EXAMPLE

Well A, 100 m depth, sensor S2: 30 °C. With a justified error bound ± 0.2 °C, the interval is [29.8, 30.2] °C. Keep the source and calibration version.

METAPHOR

A measurement needs both a return address and a ruler showing its fuzziness.

OSDU Data Platform

DEFINITION

The OSDU Data Platform provides shared data formats and services for storing, finding and accessing energy data. It links measurements to metadata and controls who can access them.

Conceptual path: record ID → authorized lookup → data + metadata

WORKED EXAMPLE

Record 17 identifies a Well A temperature log. An authorized request retrieves its 100 m, 30 °C measurement together with the curve's depth and temperature units.

METAPHOR

A shared catalog lets different laboratories find and describe materials using compatible conventions.

Open source and reproducibility

DEFINITION

Open-source licensing permits use, study, modification, and redistribution under stated terms. Reproducibility requires enough materials to repeat and evaluate a result.

Reproducible result: data + code + settings + environment

WORKED EXAMPLE

Publish allowed data (2, 4, 6), code $\text{mean}(x)$, and the environment. Another person obtains $(2+4+6)/3 = 4$. Publishing only “result = 4” cannot reproduce the calculation.

METAPHOR

Share the recipe, ingredients, oven settings, and tasting rules so others can check the cake.

Representation alignment

DEFINITION

Representation alignment connects domain vectors or tokens to language tasks. Paired examples can train a model to use those representations when predicting an answer.

A supervised objective: minimize $-\sum_i \ln P_{\theta}(\text{answer}_i \mid q_i, \text{codes}_i, \text{metadata}_i)$

WORKED EXAMPLE

For a code paired with answer “rising,” predicting probability 0.2 gives loss $-\ln(0.2) \approx 1.61$. At 0.8, loss ≈ 0.22 . Paired data teach the connection.

METAPHOR

Inventing an alphabet is easy; teaching readers what its symbols refer to takes examples and practice.

Sememe.ai framework

DEFINITION

Sememe.ai is the framework and SDK described for connecting domain models and representations to language interfaces. This well-log workflow is a proposed application.

well window → encoder → code IDs + metadata → language/model interface

WORKED EXAMPLE

A proposed design maps 512 measurements into 8 code IDs while retaining the well, depth, units, and model version.

METAPHOR

Build a translation workbench that keeps a scientific notebook beside its shorthand dictionary.

Benchmark and ablation

DEFINITION

A benchmark defines fixed evaluation tasks and metrics. An ablation changes one component to test what it contributes under a controlled comparison.

$\text{accuracy} = \text{correct} / \text{total}$; $\text{change} = \text{scorefull} - \text{scorewithout}$

WORKED EXAMPLE

On the same 50 held-out questions, the full system gets 40 right (80%). Disabling its learned-token input gives 35 (70%). The observed drop is 10 percentage points.

METAPHOR

Race on the same course, then remove one engine part to see whether it actually helped.

Forward-deployed engineering

DEFINITION

Forward-deployed engineering embeds technical work with users to define a valuable problem, integrate a solution, and measure its performance in real operations.

time saved per case = previous time – deployed time

WORKED EXAMPLE

If a verified workflow reduces a task from 30 to 10 minutes, 12 tasks save 240 minutes, or 4 hours.

METAPHOR

An engineer builds a bridge at the actual river, adjusting to its banks and traffic.

Grounded model conversation

DEFINITION

A grounded model conversation links a natural-language question to identified data and a defined computation, then reports the result with assumptions and evidence.

question → validated inputs → model/tool → result → explained answer

WORKED EXAMPLE

“Double inflow.” Starting volume 10 L, usual inflow 3 L/step, drain 2 L/step. Model predicts $10 + 2 \times 3 - 2 = 14$ L after one step, assuming no overflow.

METAPHOR

Language is the interpreter at the laboratory door; instruments and models do the measured work inside.